BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, ABBOTTABAD

INTERMEDIATE (11TH CLASS) EXAMINATION, 2023

PHYSICS

Past Paper 2023 • Paper Code: 11231

Retyped faithfully from the original board paper — same questions, no changes.

SECTION-A

Time Allowed: 20 Minutes | **Maximum Marks:** 18

Note: Attempt all questions of Section-A by filling the corresponding bubble on the MCQs RESPONSE SHEET. It is mandatory to return the attempted MCQs sheet to the Superintendent within given time.

Q1. The precision of the measurement 385,000 km is ..... km.

(A) 10
(B) 100
(C) 1000
(D) 10000

Q2. If the resultant of two vectors each of magnitude A is also a magnitude A. The angle between the vectors will be .....

(A) 45°
(B) 60°
(C) 90°
(D) 120°

Q3. The square of $\vec{A}$ is equal to .....

(A) zero
(B) $\vec{A}\times\vec{A}$
(C) $\vec{A}\cdot\vec{A}$
(D) 2A

Q4. $\vec{A}\cdot(\vec{A}\times\vec{B})$ = ...

(A) 0
(B) 1
(C) A^2B
(D) AB

Q5. Which of the following is an example of non-renewable resource?

(A) tidal energy
(B) nuclear energy
(C) geothermal energy
(D) wind energy

Q6. Kgm^2sec^{-1} is the SI unit of .....

(A) gravitational constant
(B) torque
(C) angular momentum
(D) moment of inertia

Q7. The GPS is made up of ..... satellites that orbit above the earth surface.

(A) 6
(B) 12
(C) 24
(D) none of these

Q8. The minimum velocity required to put a satellite into the orbit is called ..... velocity.

(A) terminal
(B) critical
(C) mean
(D) none of these

Q9. At terminal velocity, the acceleration of falling object is ..... m/sec^2.

(A) 9.8
(B) -9.8
(C) 1
(D) 0

Q10. A certain pipe has cross-sectional area of 0.0001 m^2 in which water is flowing. If the volume flow rate is 0.001 m^3/sec then velocity of the fluid is ..... m/sec.

(A) 0.01
(B) 0.1
(C) 10
(D) 100

Q11. The water in the tank is 10m above the leak point. The speed with which the water emerges from the leak is ..... m/sec.

(A) 14
(B) 10
(C) 196
(D) 198

Q12. Which of the following properties is not exhibited by the longitudinal waves?

(A) reflection
(B) interference
(C) diffraction
(D) polarization

Q13. The angle at which the first Bragg diffraction is observed for x-rays of wavelength 3 nm incident on a crystal of Lattice spacing of 5 nm is .....

(A) 15°
(B) 16°
(C) 17°
(D) 10°

Q14. In an isothermal change, internal energy .....

(A) decreases
(B) increases
(C) becomes zero
(D) remains constant

Q15. The dimensions of gravitational constant are .....

(A) $[M^2L^3T^{-2}]$
(B) $[ML^2T^{-2}]$
(C) $[L^3M^{-1}T^{-2}]$
(D) none

Q16. If the magnitude of the cross product of two non-zero vectors $\vec{A}$&$\vec{B}$ is AB then the dot product of $\vec{A}$&$\vec{B}$ is .....

(A) 0 (B) AB

(C) $-AB$ (D) none of these

Q17. In an isolated system, the total energy of vibrating mass and spring is .....

(A) variable (B) low

(C) high (D) constant

Q18. A 4m high tank filled with water is drilled with fair identical small holes at 1m, 1.5m, 2m and 3m from the top of the tank, the speed of efflux will be greatest from the hole at .....

(A) 1m (B) 1.5m

(C) 2m (D) 3m

SECTION-B

Time Allowed: 2 Hours 40 Minutes | **Maximum Marks:** 40

Q1. Attempt any ten of the following short questions. Each question carries 4 marks.

(i) How two forces of equal magnitude be oriented so that the magnitude of their resultant is: (i) Equal to magnitude of either of the force. (ii) Equal to twice of the magnitude of the either force.

(ii) Calculate the angle of projection for which kinetic energy at the summit is equal to one-fourth of its kinetic energy at point of projection.

(iii) When an arrow is shot from its bow, it has kinetic energy. From where does it get the energy? Discuss.

(iv) Is it possible for a person to distinguish between a raw egg and a hard boiled one by spinning on a table. Elaborate.

(v) Why does blowing over the top of a paper make the paper rise?

(vi) Show that the total kinetic energy of solid cylinder of moment of inertia about an axis through its centre of mass $MR^2/2$, is three times its rotational kinetic energy about its centre of mass i.e. K.E_T=3K.E_R

(vii) Why is a rifle barrel rifled?

(viii) A cricket ball moves past an observer from left to right, spinning counter clockwise, in which direction will the ball tend to deflect? Explain briefly.

(ix) How can you compare the masses of two bodies by observing their period of oscillation when supported by a spring?

(x) How food gets heated up in microwave oven? Why is it prohibited to use utensils made up of conductor in it.

(xi) Why does a sound wave travel faster in solids than gases?

(xii) Differentiate between real and apparent weight.

(xiii) How much work is done by a person on a suitcase carrying in his hand? Describe.

SECTION-C

Maximum Marks: 27

Note: Attempt any three of the following questions. Each question carries 9 marks.

Q2.

(i) Define elastic and inelastic collision. Derive mathematical expression for the final velocities of the elastically colliding bodies in one dimension.

(ii) Vector $\vec{F}$ having magnitude 5.5N makes 10° with x-axis and vector $\vec{r}$ with magnitude 4.3m makes 90° with x-axis. Determine the magnitudes of their dot and cross product.

Q3.

(i) What do you mean by escape velocity from earth surface? Derive an expression for the escape velocity of a body from the earth gravity. Calculate the numerical value of this velocity.

(ii) The orbital speed to launch a satellite in circular orbit above the earth surface is 7.4×10^3m/sec. Calculate the distance of the orbit from earth surface.

Q4.

(i) What is SHM? Show that energy is conserved in case of SHM.

(ii) A body of mass m suspended from a spring with force constant k, vibrates with f_1. When its length is cut into half and the same body is suspended from one of the halves, the frequency is f_2. Show that $f_1/f_2 = 1/\sqrt{2}$

Q5.

(i) Define molar specific at constant volume and constant pressure of a gas. Show that for n moles of an ideal gas C_p − C_v = R.

(ii) A diffraction grating has 5000 lines per centimeter. At what angle does the second order spectrum of the sodium yellow light of wavelength 589nm occur?